

Introduction

In recent years, consumers have gained greater means of self production through the major breakthroughs in the performance and reliability of consumer-grade 3D printers. Yet, even though there has been a huge growth in urbanization, there are many counties within the United States that are still lacking access to grocery stores (Bonderson, 2023). With the recent growth of the consumer Fused Filament Fabrication (FFF) 3D printing industry, many remote regions, such as those without grocery stores, have gained access to greater manufacturing capabilities. FFF 3D printers can be utilized to jump-start a consumer based production line allowing the production of innovative, creative solutions to day-to-day problems. For instance, researchers for the international journal of advanced manufacturing technology found that the Selective Laser Melting (SLM) 3D printing process was an effective method of creating high quality microfluidic molds (Colin et al., 2021). This study will test whether 3D printing could be utilized to fill the gap in access to markets that carry food container products within rural areas.

FFF 3D printing utilizes many different types of plastics while printing, with varying levels of safety. As a result of the often dubious information surrounding printing assorted filaments, it can be difficult to determine whether a certain type of plastic is food safe. Research has been conducted regarding 3D printer emissions during printing, documenting significant numbers of nanoparticles emitting when printing using Acrylonitrile Butadiene Styrene (ABS) and a less significant finding when using Polylactic Acid (PLA) (Mendes et al., 2017). While emissions may give some insight into the safety of plastics during printing, less research, if any, has been conducted to link plastics as definitively safe to use when ingesting out of 3D printed containers-similar to cups.

One major issue with 3D printed containers, especially those that store liquids, is the high likelihood of leaching microplastics from the container into the contents of which the vessel is storing. On the topic of microplastics found in water, plastics such as polyethylene, polypropylene, and polystyrene were recorded as the greatest polymer frequency in a study conducted by researchers in the Netherlands (Koelmans et al., 2019). As opposed to commonplace research focused on analyzing the microplastic contents of varying types of plastics, this research article will focus on analyzing the relationship between assorted plastics and the counts of microplastics found during microscopic analysis providing insight using different factors such as temperature of the water, time the water spent in the carrier, and layer height. In order to collect the counts of microplastics, a testing method called “microscopic analysis” will be utilized to put the sampling liquid, water, into a grid to search through an oil-based microscope and individually log each particulate. The current hypothesis states that more microplastics will be observed in ASA as it has been considered a toxic filament and most microplastics will have a direct relationship with the infill percentage and temperature of the print.

Literature Review

Historically, sources surrounding the topic of microplastics within 3D printed objects had three distinct themes: emissions from the 3D printing FDM process, microplastics that are found within sources of water, and the safety of human consumption of microplastics. Consequently, the gap of my research will be surrounded by the limitations of these three themes.

3D printer emissions

Within the field of consumer 3D printing, it has long been known that the emissions released from the printing process reduce IAQ (indoor air quality), producing particulate matter

that can be dangerous to breathe. The process of FDM 3D printing is known to be melting layer upon layer of plastic in order to result in one cohesive assembly of layers (Niosh, 2023). During the process of printing, one study conducted on a desktop 3D printer reported releasing particulate matter that was “7.8 to 10.5 nm in regular printing” (Mendes et al., 2017, p. 104) which are classified as micro-emissions similar to microplastics. However, there are various factors that play into the production of particulate matter emissions such as infill density and printing temperature. In a study conducted by Khaki and colleagues (2021) focused on 3D printer emissions within the home setting, infill density was found inversely correlating with printer emissions while printing temperature directly correlated with printer emissions. Although, one of the most significant factors for printer emissions can be attributed to filament type where PLA has less ultrafine particulate matter emissions than ABS and PETG, but filaments that contain particle additives such as metal infused filaments increases both ultrafine and fine particulate matter emissions (Hill et al., 2022). With all of the toxic emissions released from 3D printing, it is important to discuss the unique benefits of 3D printing; a study utilizing LCD stereolithography printing, a process in which resin is solidified with light, highlights the ease of use in creating complex geometry using a relatively cheap 3D printer (Colin et al., 2021). As both Khaki and colleagues (2021) and Hill and associates (2022) tested the effects of variable factors such as filament type and printing settings they would be likely to agree that the emissions can vary greatly based on the parameters used when printing. Based on the later findings of Mendes and colleagues (2017) in testing ABS and PLA emissions during printing, it is likely that Mendes and associates (2017) would agree with Hill and colleagues (2022) that PLA can be considered less emissive than ABS. However, one major limitation of these studies

is the lack of discussion of potential applications of 3D printing technology in printing food safe containers especially in regards to microplastics.

Microplastics within water

Separately from particle emissions, small microparticles made from plastics often seep into water which can pose questions of quantity, size, and polymer of the microplastics. To give a more formal definition of microplastics, Akbulut and associates (2024) defines microplastics as “Plastic particles <5 mm in size” (p. 26). Considering the size of the particles within the water, it is also pertinent to the discussion to understand the common polymers of plastics within water. Previous tests for microplastics within water have found that the frequency of polymer types can be considered by the following comparison, “PE $\approx$ PP > PS > PVC > PET” (Koelmans et al., 2019, p. 419-420). The study later explained the relationship to the global demand of plastic and polymer density as the former explains the lack of PVC and PET found at the surface of the water and the latter describes the overall trends of plastics found in the water (Koelmans et al., 2019). One explanation for the entrance of microplastic polymers into water is the degradation of biodegradable plastics. Apicella and colleagues (2024) point out that plastics with a higher crystallinity degree will have slower degradation rates. Where high degradation rates mean the plastic will quickly lose integrity, and low degradation rates correlate with slower loss of integrity. As a result, both plastics that have a high crystallinity degree and low crystallinity degree will be less likely to be commonly reported due to the rate at which they break down. Contrastingly, tap water and bottled water could be less exposed to environmental microplastic contamination and degradation as they are often rigorously tested and purified for drinking safety before consumer usage. Moreover, between tap water and bottled water, tap water is assumed to be less likely to contain significant microplastics due to WWTP (wastewater treatment plants)

removing more than 99% of microplastics within a German WWTP (Lau et al., 2024, p. 288). As a result of the minimal microplastics found within tap water, bottled water will have more microplastics in relation to tap water (Koelmans et al., 2019). Furthermore, according to researchers at Wageningen University PET and PU plastic are the most common polymer types within samples of bottled water (Nizamali et al., 2023). Koelmans and associates (2019) and Lau and colleagues (2024) produced similar findings on tap water being clean from most microplastics, on the whole. However, Apicella and associates (2024) would argue that Koelmans and colleagues (2019) may not have related the common polymers of microplastics correctly by missing the degradable plastics, such as PLA and PBS. In recognizing the increased microplastic concentration within bottled waters, there could be a correlation between 3D printed plastic bottles and an increase in microplastics found within sampled water from the printed container.

Microplastic consumption

In discussing microplastics, it is important to note the impact and safety of microplastics on the human body. To establish the expected amount of microplastics a consumer would intake, a study was conducted that found MP (microplastic) exposure as between “18720-73840 MPs/year” (Akbulut et al., 2024, p. 26) for the average consumer. However, different types of microplastics have adverse effects on the human body’s systems. Čurlej and associates (2023), citing a study on microplastics within consumer food containers, mentioned that “Fadare et al.(2020) concluded, that plastic particles of smaller size can lead to more pronounced toxic effects. Furthermore, irregularly shaped particles, compared to uniformly shaped particles, are retained for longer periods in the body” (p. 3). More specifically as microplastics decrease in size, oxidative stress and depolarization increase (Jayavel et al., 2024). Importantly, oxidative

stress is known to increase the likelihood of various diseases as a result of damage to cells, tissues, and organs within the human body. To that point, Jayavel and colleagues (2024) continues that microplastics have been proven detrimental to bodily functions specifically leading to inflammation, abdominal pain, bloating, nausea, vomiting, and neurotoxicity. Wang and associates (2020) adds that PS microplastics can affect amino acid, bile acid, and lipid metabolism leading to their respective metabolism disorder. Likewise, one study focusing on microplastics and human health found that, certain types of microplastics, such as polyethylene terephthalate (PET), polystyrene (PS), and polyvinyl chloride (PVC), harm cell viability and trigger inflammatory genes by unleashing compounds that can increase risk of cancer (Damaj et al., 2024, p. 49). Both Čurlej and colleagues (2023) and Jayavel and associates (2024) found that smaller microplastics have an inverse association with health, particularly mentioning the toxic effects of consuming microplastics. Jayavel and colleagues (2024), Wang and associates (2020), and Damaj and colleagues (2024), are all likely to cite various health concerns surrounding microplastics with Jayavel and associates (2024) and Damaj and colleagues (2024) citing inflammation as a significant result of microplastics. Considering the dangers that come from microplastics, especially smaller variants, it is important to minimize the consumption of microplastics in day-to-day life.

Conclusion

While research conducted on microplastics has yielded information on generic bottled water, there has not been any discussion of 3D printed containers and the quantitative data concerning count and size of the microplastics in such containers. Furthermore, testing for 3D printer emissions centered mainly around emissions into the air as opposed to microplastics trapped into the print that can seep into liquids, such as water, and contaminate the contents held

within a 3D printed container. As a result of these shortcomings, this study plans to utilize quantitative data to analyze the impacts of 3D printed cup geometries on the amount of microplastics found within the sampled water. Thus, this study will aim to answer the question: What is the impact of 3D printing cup-shaped geometry on the presence of microplastics within sampled waters?

Methodology

The method of this study aims to respond to the following research question: what is the impact of 3D-printed materials on microplastic contamination in water? Before conducting research, it was hypothesized that more microplastics will be found in filaments with a higher printing temperature and speed but the polymer of filament would have a greater total impact on microplastics present.

Visual microscopy, also known as microscopy, involves the visual analysis of miniscule matter through a microscope (Karakulum & Krishnan, 2022). More specifically, samples can be classified and counted using the process to magnify samples on microscopy slides (Karakulum & Krishnan, 2022). As this study is focused on explaining the general presence of microplastics within different water samples, the accuracy yielded from microscopy can be used to establish statistically significant trends. However, this study will not utilize processes like FTIR (Fourier Transform Infrared) spectroscopy that require institutional facilities as it aims to conduct research in a high school laboratory. Consequently, this study will choose only visual microscopy as it's method to fully explain the relationship between 3D printed plastics and concentration of microplastics. In order to comprehensively explain trends in microplastic contamination due to close proximity with 3D printed cups, the data found on quantitative data on microplastic count, polymer, and size will be used in combination with qualitative data such as microplastic shape.

This study will collect a majority of its data through the utilization of light microscopy allowing for understanding of particles within the sampled waters. Data that is considered eligible will be centered around microplastics and trial variables; specifically, data will only be included if it pertains to the direct impact of microplastics found within the sample or contains information on the microplastics found within the sample. Only focusing on properties and direct contributors to the microplastics in the sampled waters assures that the data collected will pertain to the research question thus minimizing extraneous information and data trends. Data will be gathered during the 2024-2025 time period and will be analyzed through the use of computer software, primarily imageJ. Due to this, only one sample is required from each polymer's cup as computer imaging tools have a high degree of accuracy. In regards to visual microscopy, an oil immersion microscope will be leveraged to view the samples and data will be recorded through digital photography. All data will be organized through digital software and graphically represented to illustrate potential trends.

The experimentation process will make use of a regular 40-400x scientific microscope, 3D-printed cups from a consumer-grade 3D printer, digital camera, and standard laboratory equipment. Before the trial can begin, one cup will be printed in PLA at 220°C with 10% cubic infill and 100% part cooling, but two other cups will vary the temperature by 10°C, 210°C and 230°C, one cup will be printed with 75% cubic infill and another will be printed at 50% part cooling each keeping the other settings the same. For the cups differing by filament type, two cups will be printed at the recommended temperature and 100% part cooling, but one cup will be printed with 10% cubic infill and the other cup will be printed at 75% cubic infill. Each cup's print settings aside from major factors will be taken from Bambulab's Bambu Studio with filaments that are not Bambulab brand filament using the "generic" presets. All testing will be

conducted as a comparison between the sampled water from printed cups and control water from a ceramic or glass container.

After printing the cup, pour 300mL of water into the cup and set a timer for five minutes to model a realistic use scenario. Then, take a sample of the water using a dropper and positioning the dropper as close to the side of the cup as possible. Afterwards, drop the sample onto a microscopy slide and focus the microscope so that it is in 40x zoom and small, inorganic washer shapes are visible. Next, using a digital camera or phone camera, center the frame down the microscope lens and take multiple photos of the sample, shifting the position between each photo so that the same section of water is not being captured more than once.

Upon completion of the experiment, data analysis can be conducted through the use of computer software. Data collected from visual microscopy will be analyzed using ImageJ yielding information on particle size, shape, and relative count. In order to conduct this analysis, each image will be cropped to only the section of the area that is shown under the microscope. Following this, automatically adjust the image with the brightness/contrast tool and use the enhance contrast tool with equalize histogram checked off. Then, the image will be changed into 8-bit typing. After this, use the subtract background process with a rolling ball radius of 50 and the light background setting turned on. Next, the image will be adjusted with the threshold tool automatically deciding the threshold based on the image and the dark background setting turned off. Finally, the particles will be analyzed with pixel sizes ranging from 20 to 99,999, circularity ranging from 0.20-1.00, showing outlines, including holes, displaying results, and summarizing the findings.

This study will use exploratory data analysis (EDA) employing graphical techniques to maximize insight into a data set and extract important variables (*1.1.1. What is EDA?*, 2019).

Using EDA, graphical constructs will be created to visualize and produce statistically significant trends. EDA is effective in establishing the relationships and patterns from the data collected allowing correlations in microplastics to be determined from the testing.

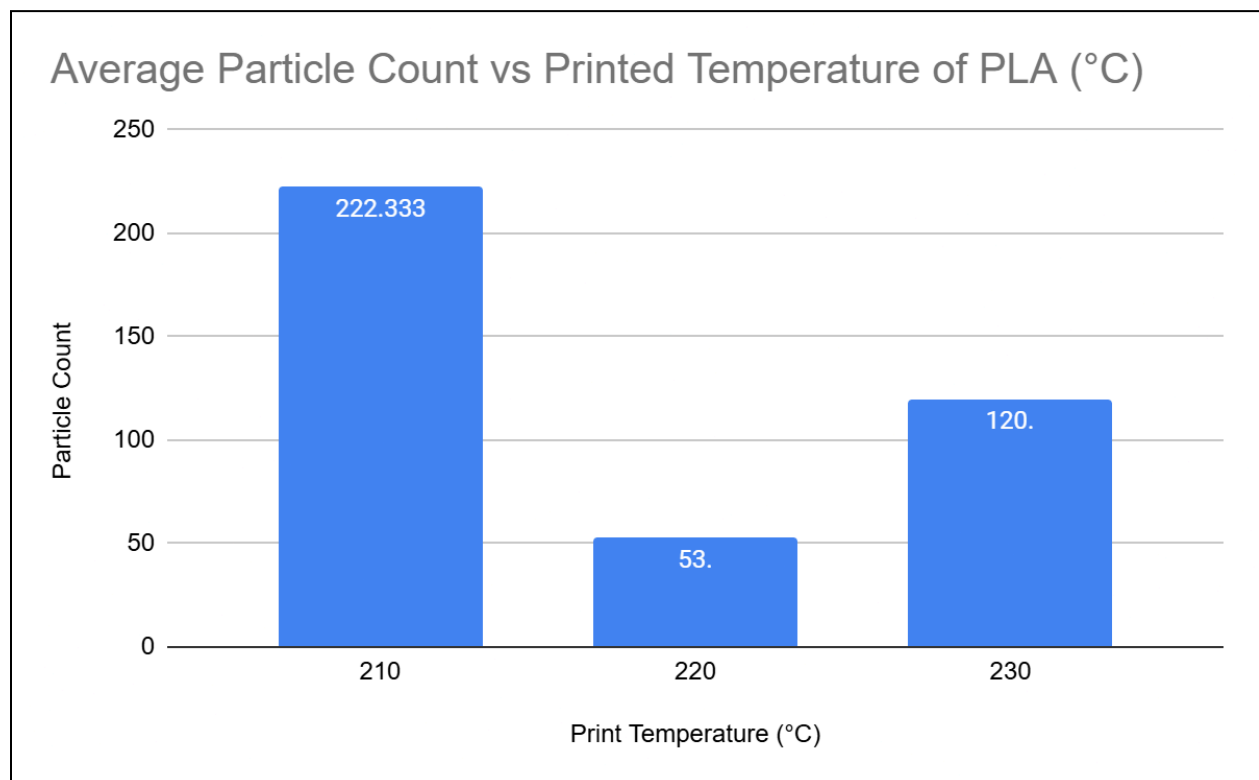
Results

In the analysis of the images, multiple trends were found revolving around the materials and properties of the 3D printed cups that could help to answer the question: what is the impact of 3D printing cup-shaped geometry on the presence of microplastics within sampled waters?

Comparisons between printing temperature of PLA cups and particle count neither indicated a positive nor a negative trend rather indicating a deficit in particle counts when the cup was printed at the recommended temperature, 220°C, as seen in Figure 1.

Figure 1

PLA Samples Differing based on Temperature



This is further supported by the t-test regarding figure 1 that failed to reject the null hypothesis because of a p-value > 0.05 signifying that there is not enough evidence to conclude a positive relationship between temperature and particle count as shown in Figure 2.

Figure 2

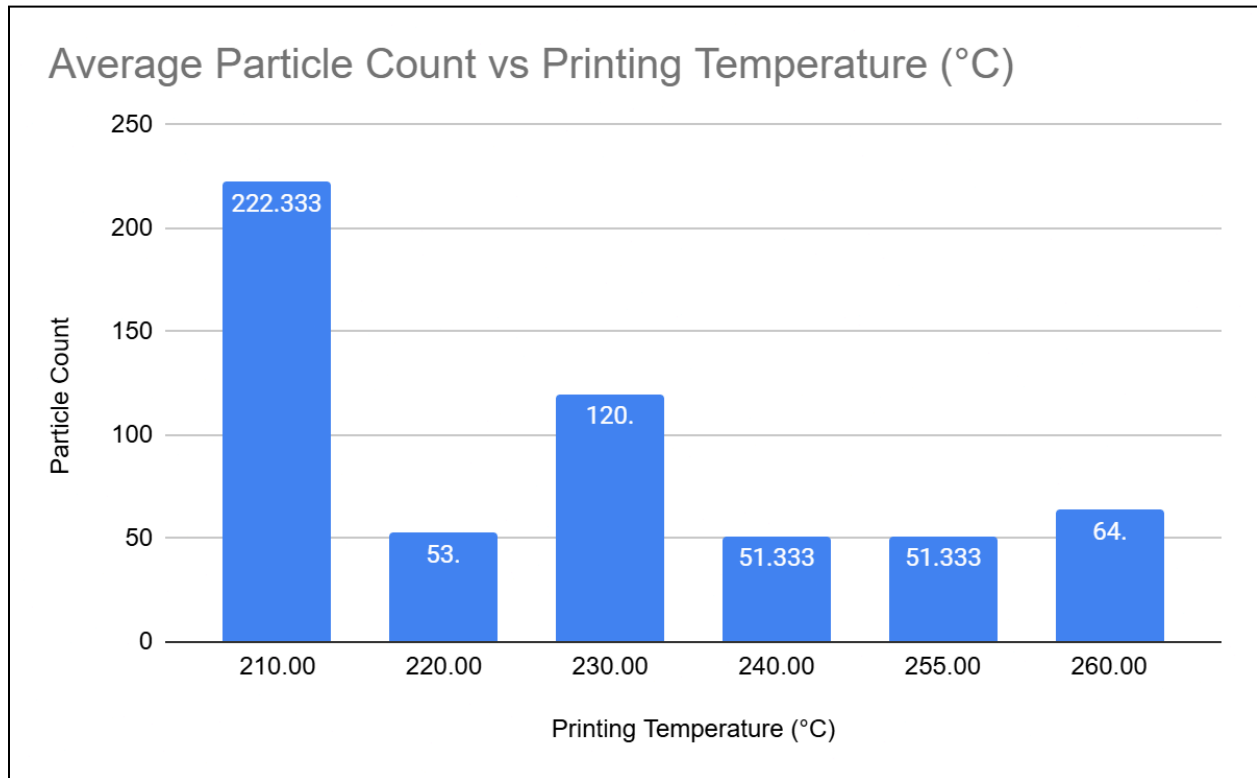
T-test Analyzing Trends Between Printing Temperature of PLA Cups and Particle Count

Test Statistic	Degrees of Freedom	P-Value	Correlation Coefficient
-1.532	7	0.915	-0.501

Furthermore, the t-test seems to be more fitting of a negative relationship between temperature and particle count although still not to a significant quantity. Contrary to Khaki and colleagues (2021) findings, printing temperature was not found to have a direct relationship with particle presence; in fact, when looking into general trends between printing temperature and particle counts, there is a clear negative correlation as the temperature increases and the particle counts decrease which can be observed in Figure 3.

Figure 3

Particle Count of All Printed Samples According to Temperature



After running a t-test, the data analyzed in Figure 3 disproved the null hypothesis with a p-value < 0.05 suggesting a negative relationship between temperature and particle count. The t-test also provided evidence for the negative trend with a test statistic of -2.966 as shown in Figure 4.

Figure 4

T-test Analyzing Trends Between Printing Temperature and Particle Count

Test Statistic	Degrees of Freedom	P-Value	Correlation Coefficient
-2.966	16	0.005	0.596

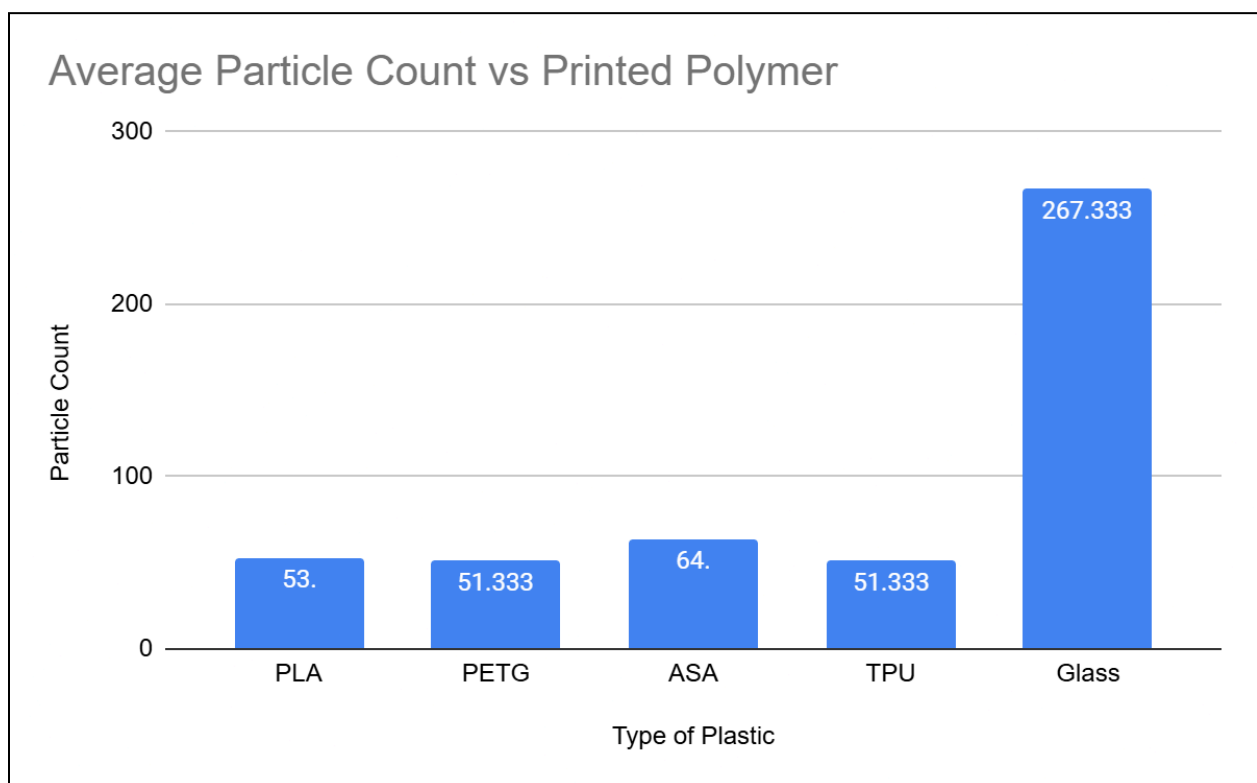
The significance of this data illustrates that even though it was originally suspected that printing temperature would directly correlate with particle count, the data suggests just the

opposite: as printing temperature increases, the particle count will decrease according to data collected from all of the printed samples.

Originally, the hypothesis suspected that the different types of polymers would be the factor with the greatest impact on the amount of particles found, but, upon completion of data analysis, it would seem that there is little correlation between polymers as suggested by Figure 5.

Figure 5

Average Particle Count according to Material Sampled



After conducting an ANOVA test it was found that the null hypothesis was rejected because the $p\text{-value} < 0.05$ meaning there was enough evidence to conclude a significant difference between the mean values of particle count according to the polymer being printed as shown by Figure 6.

Figure 6

ANOVA Test Comparing Printed Polymer and Particle Count

Test Statistic	P-value
35.864	0.000007

However, it can also be seen in Figure 5 that glass had an outlier amount of particles. In testing, I believe this to have been caused by a reusage of the same sampling beaker which led to accumulation of particles from the multiple days of testing. Another explanation for the greatly increased observation of particles in the glass samples could be the time waited until sampling. During sampling, it was five minutes before any of the printed cups' water were sampled, but the glass sample was taken almost immediately after filling the beaker which could lead to more of the particles being gathered towards the top of the container, where the data was being collected.

Furthering the discussions of the plastic polymer being printed, the averages of the total area of the particles and the percent area covered by the particles show a large difference based on the material being analyzed. This can be seen in Figure 7 and Figure 8.

Figure 7

Total Particle Area According to Material

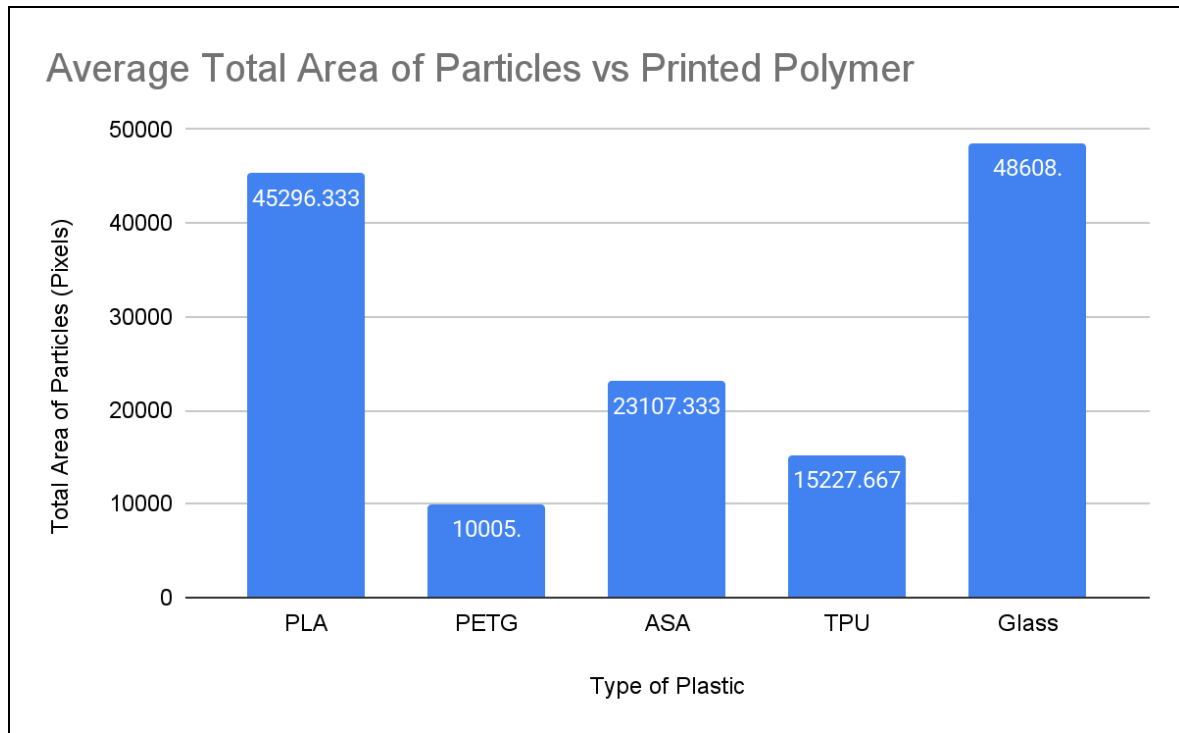
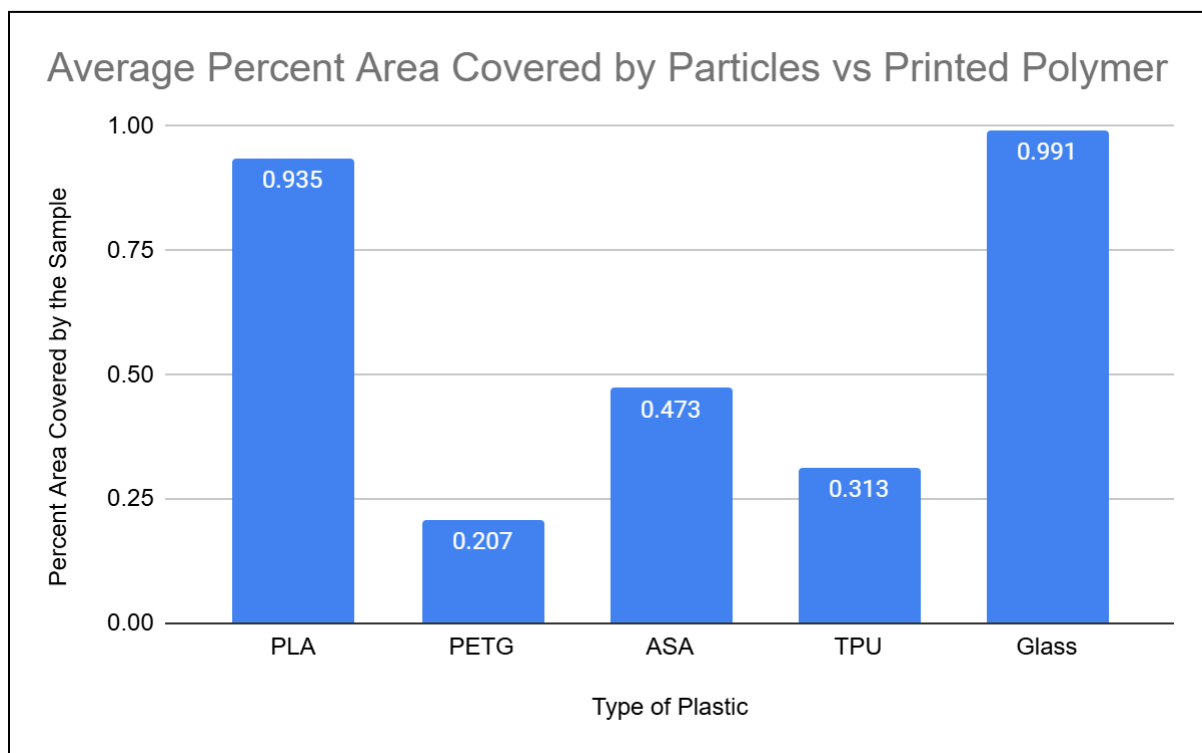


Figure 8

Percent of Image Covered by Particles According to Material



While Hill and associates (2022) found that PLA had the least particulate matter emissions when compared to ABS and PETG, Figure 7 and Figure 8 both support the idea that PLA produces the greatest particle presence among the printed samples. Furthermore, for both Figure 7 and Figure 8, the null hypothesis could be rejected from ANOVA testing with enough evidence to conclude a significant difference between the mean values of total particle area and percent area covered according to the polymer being printed as displayed in Figure 9.

Figure 9

ANOVA Test Results Comparing Printed Polymer to both Total Area and Percent Area

Data Tested	Test Statistic	P-value
Printed Polymer vs Total Area	4.974	0.018
Printed Polymer vs Percent Area	4.778	0.02

Within the initial hypothesis, it was also suspected that infill would have a constant positive effect on the amount of particles observed. However, varying trends were identified based on polymer as well as some materials having negligible changes in particles based on infill percentage. The general trend did not follow the outline of infill density given by Khaki and colleagues (2021), but some types of filaments did adhere to the trend suggested by Khaki and colleagues (2021). While not statistically significant as shown in Figure 10, the general trend observed by increasing infill can be characterized best as positive which is shown in Figure 11.

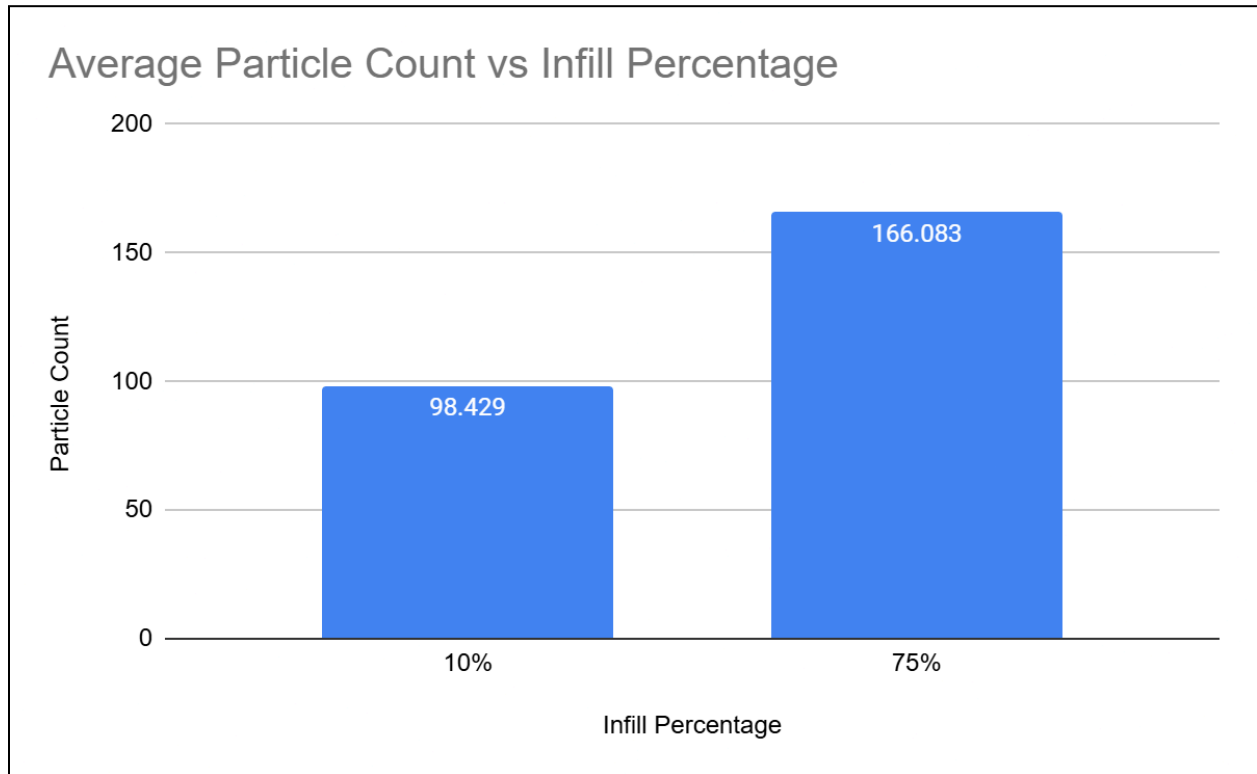
Figure 10

T-test Analyzing Trends Between Particle Count and Infill Percentage of All Printed Cups

Test Statistic	Degrees of Freedom	P-Value	Correlation Coefficient
1.565	31	0.0638	0.271

Figure 11

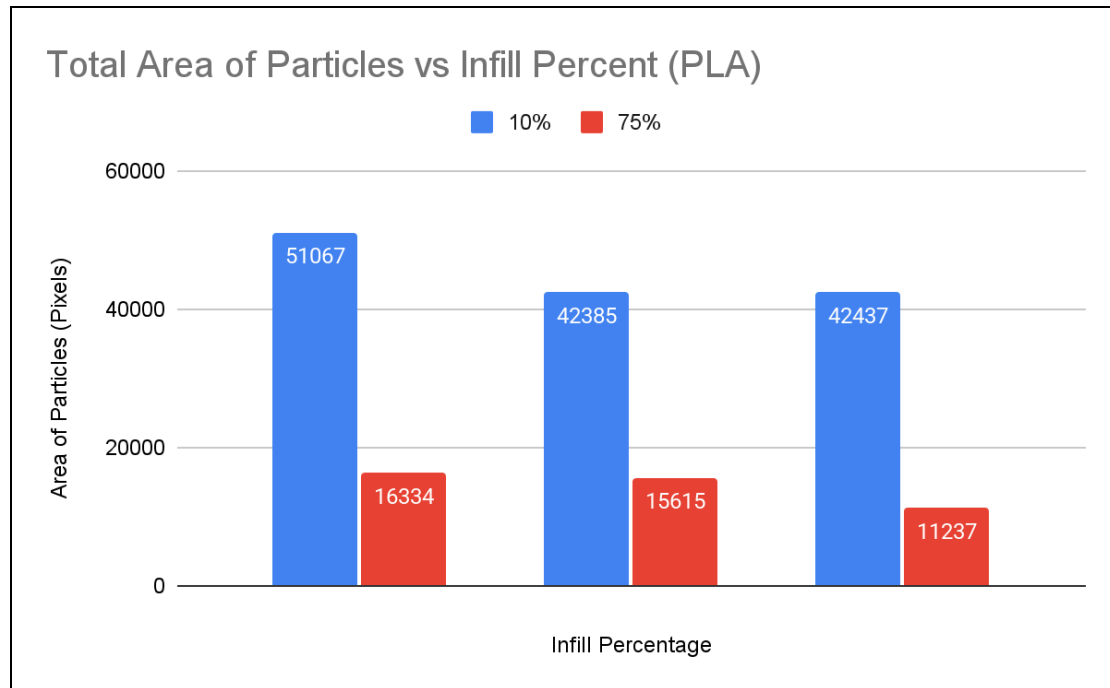
All Printed Sample's Particle Counts Analyzed According to Infill Percentage



In specific filaments, more statistically significant trends regarding infill percentage can be observed as in the case of PLA and TPU. For example, PLA particle count displayed a negative correlation due to the area of particles decreasing with infill percentage increasing as shown in Figure 12.

Figure 12

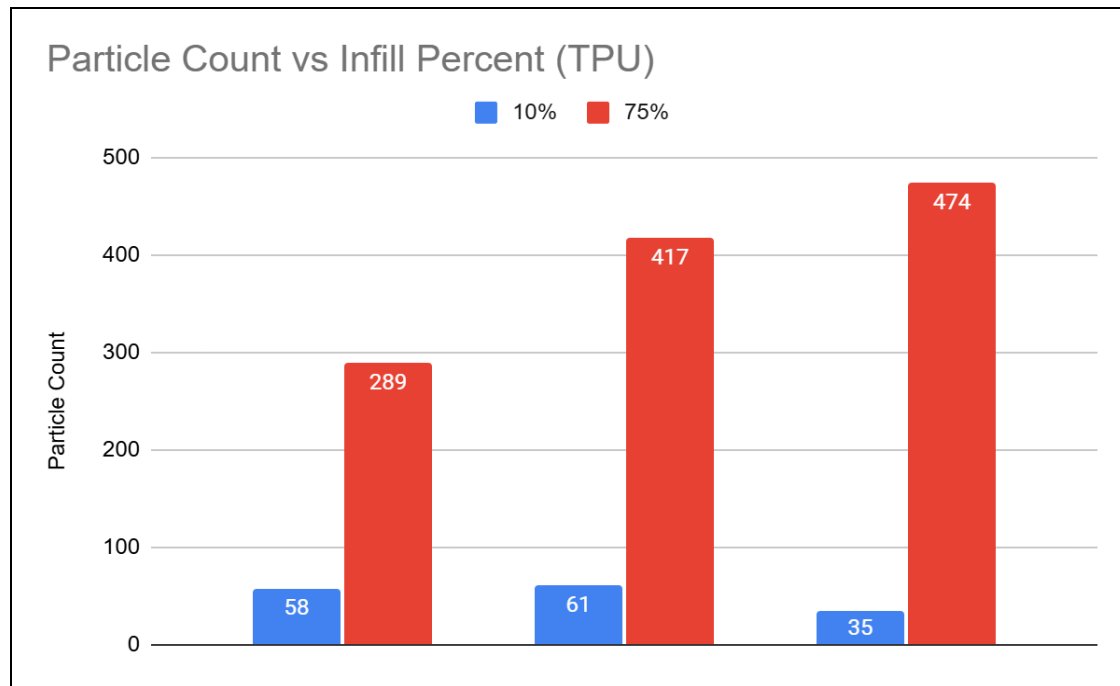
PLA Printed Cups' Particle Area Differing by Infill Percent



Contrastingly, the infill percentage of TPU had a direct correlation with the amount of particles illustrated in Figure 13 demonstrating an increase in particle count as infill percent increases.

Figure 13

TPU Printed Cups' Particle Count Differing by Infill Percent



As a result of a p-value < 0.05 , there is enough evidence to conclude a negative relationship between infill and total area covered by particles for PLA filaments as shown in Figure 14; however, there is also enough evidence to conclude a positive relationship between infill and particle count for TPU filaments because of the p-value being less than 0.05 as shown in Figure 14.

Figure 14

T-test Results for PLA and TPU Filaments Differing by Infill Percent

Data Tested	Test Statistic	Degrees of Freedom	P-Value	Correlation Coefficient
Total Area vs Infill Percent (PLA)	-9.376	4	0.0004	-0.978
Particle Count vs Infill Percent (TPU)	6.183	4	0.002	0.951

Another factor in printing, part cooling, displayed an interesting positive relationship with total particle area and percent area covered by particles. This trend can be seen in Figure 15

and Figure 16 as the increase in part cooling percentage from 50% to 100% shows an increase in both total area as well as percent area covered by the particles.

Figure 15

Area Covered by Particles within PLA Printed Cups of Varying Part Cooling Percentages

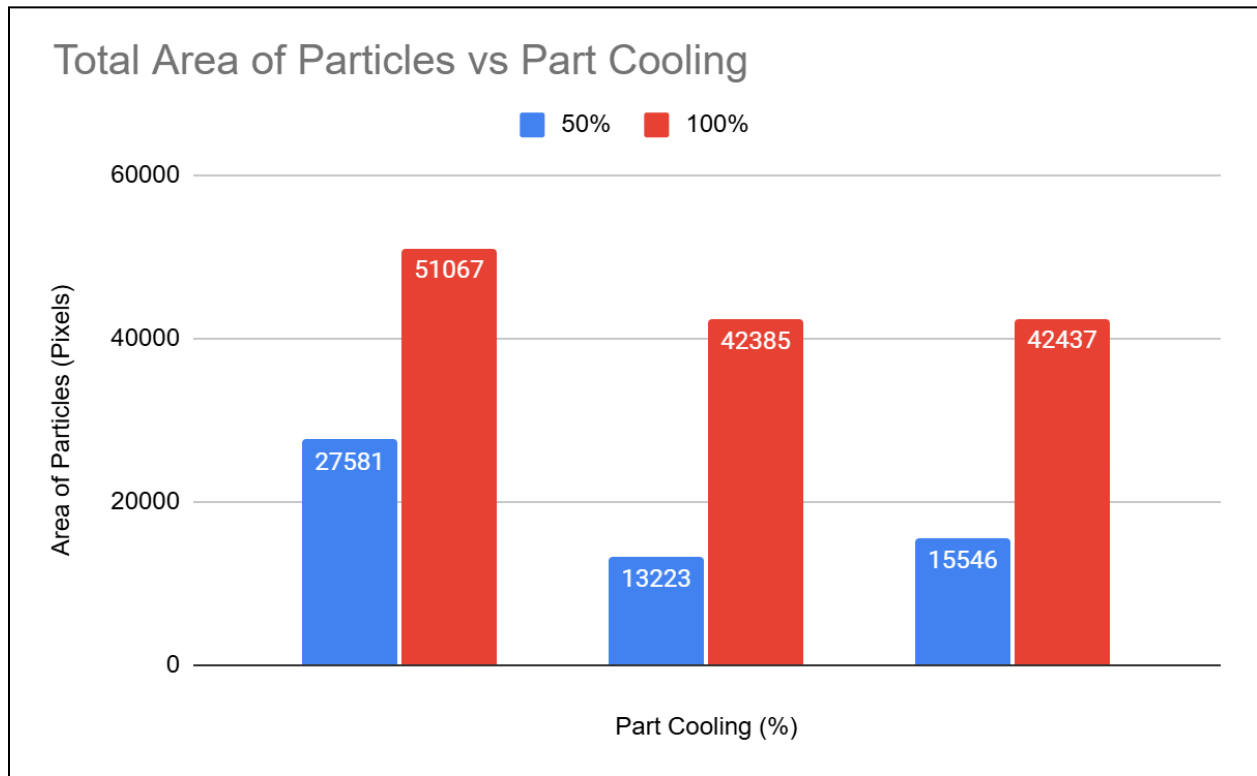
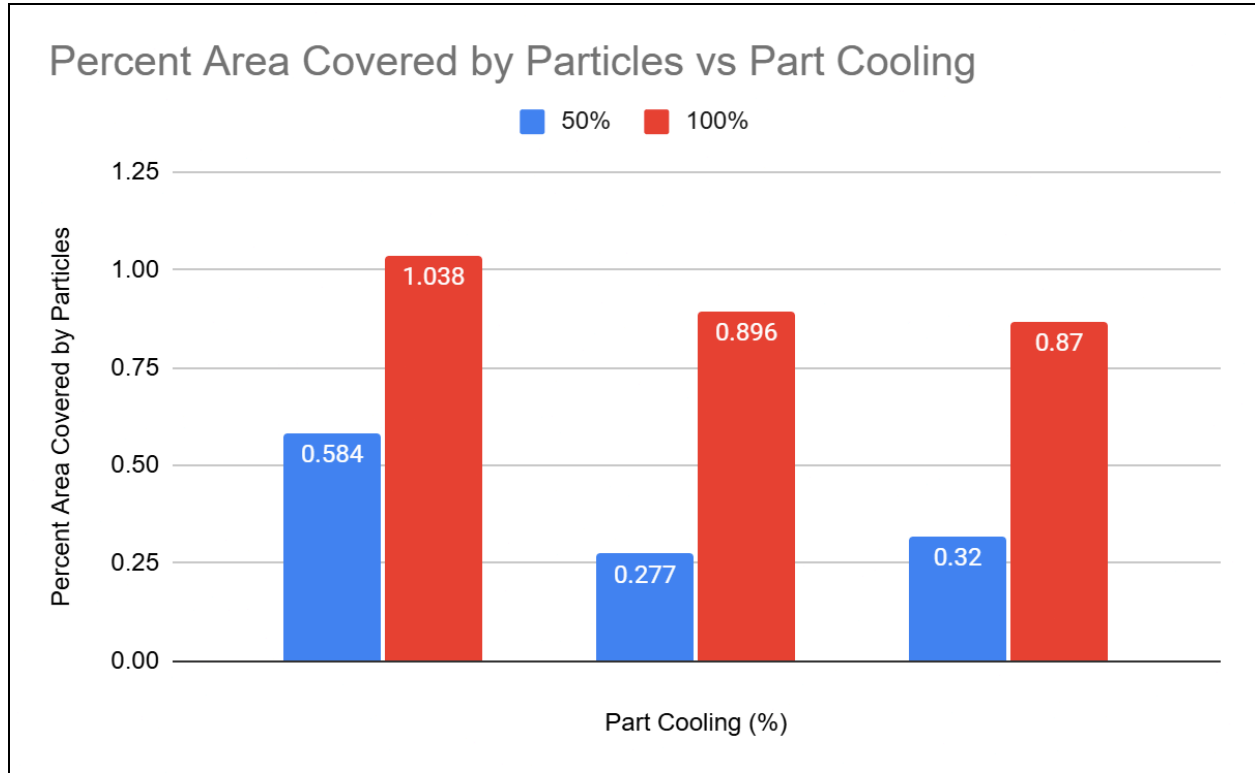


Figure 16

Percent of Area Covered within PLA Printed Cups of Varying Part Cooling Percentages



The data represented by Figure 15 and Figure 16 both reject their null hypothesis, respectively. Meaning that there is enough evidence to support a positive relationship between part cooling and total area as well as percent area covered by the particles which is further backed up by Figure 17.

Figure 17

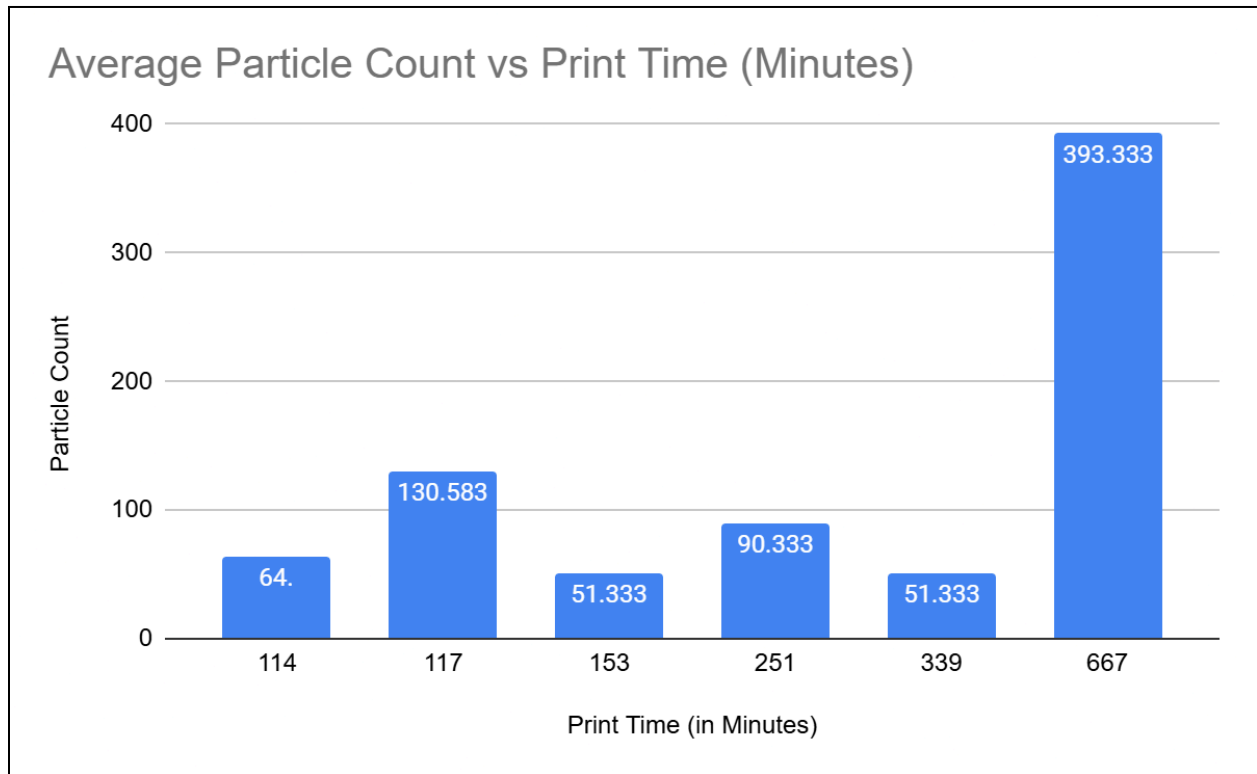
T-test Analysis of Part Cooling's Relationship with both Total Area and Percent Area

Data Tested	Test Statistic	Degrees of Freedom	P-Value	Correlation Coefficient
Total Area vs Part Cooling	4.999	4	0.004	0.928
Particle Count vs Part Cooling	4.952	4	0.004	0.927

Additionally, all the print times of each cup were compared to the particle properties observed after analysis yielding trends regarding the time spent in printing. The particle count, for example, displayed an increase as print time continued, shown in Figure 18.

Figure 18

All Printed Cups' Particle Count According to Time Spent 3D Printing



The t-test conducted between particle count and print time yielded a p-value < 0.05 meaning that the null hypothesis can be rejected and that there is enough evidence to conclude a positive relationship between print time and particle count which is supported by Figure 19.

Figure 19

T-test Analysis of Particle Count and Print Times Across All Printed Cups

Test Statistic	Degrees of Freedom	P-Value	Correlation Coefficient
3.772	31	0.000343	0.561

In conclusion, five factors were shown to have a significant impact on the presence of particles within the tested water samples. As a general factor, temperature was identified as being a negative relationship as shown in Figure 3; however, in testing PLA filament where the factors were more isolated, temperature was found to have a negligible trend in effecting particle count in more explicitly tested conditions. Although printed polymer was expected to play a major role in the presence of particles, in testing the only trend identified heavily relied on the outlying glass sample, which was more-than-likely faulty. However, PETG stood out as having far less particles present in agreement with the general consensus that PETG is a food safe filament. One unexpected trend that arose was the difference in the impact that infill percentage had on particle presence when printed in different filaments. While PLA had a negative trend between infill percentage and particle presence, TPU had a positive trend that suggested an increase in infill would lead to an increase in particle presence. Part cooling, which was only analyzed in regards to PLA printed cups, displayed a direct correlation between particle presence and the percent of part cooling being utilized while printing. The data also points towards print time having a positive relationship with particle presence although the data is taken from all different kinds of printed polymer samples so the data could be more indicative of interpolymer relations.

Discussion

The objective of this study was to find the extent to which 3D printing cups had an impact on the microplastic quantities in water taken from the cup. Throughout discussions of microplastic presence it will be assumed that changes in particle presence will correspond to microplastic presence as the varying factors should have no impact on the particles observed in the water outside of microplastics. While it was initially hypothesized that ASA has more microplastics than other filaments, the results of the study indicated that PLA has the greatest

microplastic presence. Furthermore, the original framework suspected that both temperature and infill density would have a positive correlation with microplastic quantity, but the general trend for temperature suggested a negative correlation with microplastic quantity in disagreement with the framework, and the general trend for infill density suggested a positive correlation with microplastic quantity in agreement with the framework.

There were multiple factors identified by the results that serve to answer the question: what is the impact of 3D printing cup-shaped geometry on the presence of microplastics within sampled waters? However, overall, the findings support the idea that it is important to inspect each individual variable factor when determining the microplastic presence. For example, even within analyzing the same factor, infill percentage, the differing positive and negative trends exemplify the reason that individual relationships should be analyzed. In a study conducted by Khaki et al. (2021), it was found that “as infill density increased, overall PM emissions decreased” (p. 9) which suggests that this study should observe a decrease in microplastic presence as infill density increases. Of course, even though the general trend throughout most of the filaments is statistically insignificant, it is somewhat indicative of a direct correlation that opposes the findings shown by Khaki et al. (2021). Contrastingly, the statistically significant data for TPU based infill testing displayed a trend in agreement with the findings presented by Khaki et al. (2021) due to a trend observed where an increase in infill percentage decreased the presence of microplastics.

The type of plastic polymer that was printed also had an observable effect on the presence of microplastics within the water samples. The findings support the idea that PLA plastic had the greatest amount of microplastics observed with ASA, TPU, and PETG following, respectively, as shown by Figure 7 and Figure 8. This did not follow the trends outlined in this study’s initial

hypothesis as it was expected that a filament chemically similar to ASA filament, ABS filament, would exhibit the greatest microplastic presence, but the results indicated that PLA actually displayed the greatest microplastic presence. This trend was found in disagreement with a study by Finnegan et al. (2023) which had claimed that “ABS black BASF filament resulted in the highest emission of particulates” (p. 11). Although, in a study by Apicella et al. (2024) the degradation of plastic polymers yielded results that PLA had the greatest rate of degradation in water when compared to PBS and PBS/PLA 70/30 plastics. Potentially creating reasoning for the discrepancy between greater observed microplastics in PLA when tested in water even if it was concluded that ABS would have greater particle emissions. It can be concluded from Figure 9 that the polymer printed had a significant effect on microplastic presence which means that as a factor, the printed polymer has an impact on the microplastics present within 3D printed cups.

Another factor that was hypothesized over was temperature. Temperature was found to have a relatively negligible effect on microplastic presence when isolated from changing polymers, but when the effect was compared to all of the polymers printed at different temperatures, a negative association was found between the printing temperature and microplastic presence. The insignificant data found in Figure 2 and shown by Figure 1 demonstrates the negligible effect that changing print temperature had on microplastic presence while the significant data found in Figure 4 and shown by Figure 3 clearly illustrates the negative association between the printing temperature and microplastic presence. However, the validity of this trend could be called into question due to the range of filament polymers analyzed in order to construct the t-test and graph that represents the negative correlation. In the study by Khaki et al. (2021), the researchers concluded that “the higher the extrusion temperature, the greater the overall PM emissions for both filament types” (p. 10). Thus the results found in this study do not

concur with the findings of Khaki et al. (2021) since this study found that as the extrusion temperature increases there is less microplastic presence. However, this could be explained if you consider the particulate matter (PM) emissions produced while printing to limit the amount of microplastics that can be emitted after printing. That is to say, there could be a trend that as PM emissions increase, the number of microplastics present decreases.

The limitations of this study include the improper glass control sampling, human-based random sampling under the microscope, and the multiple day study-length. As previously mentioned, during the sampling for the glass control sample, improper precautions were taken and the method of sampling outlined by the methodology section was not properly followed. Due to this complication, the control sample was not able to be used to compare the amount of particles added to the water sample by the cups as opposed to the general amount of particles observable in the tap water. In addition, during the digital image collection process, the image under the microscope could vary significantly based on where the researcher decided to take an image of the sample. For example, certain sections would be almost bare of particles with others being far more apparently visible. This uncertainty in data collection cannot directly be attributed to any specific abnormalities, but it could be important to implement more samples to reduce the human error from biased digital image collection. Finally, the multiple-day study length meant that the tap water's concentration of particles could vary day-to-day resulting in the creation of potentially false trends in the data arising. For future studies, it would be important to keep the data sampling standardized in addition to adding greater sample size and conducting the test over the span of one day to mitigate these credibility issues.

Implications and Conclusion

In order to investigate the impacts of 3D printed plastics on the amount of microplastics observed in water, this study conducted research that yielded results on various confounding factors that illustrate a greater picture of the initial inquiry. In doing so, this study refuted its initial claim that the most microplastics will be observed in ASA as well as the claim that temperature has a direct correlation with the number of observed microplastics. Although, the initial hypothesis regarding infill percentage was discovered to depend greatly based on filament polymer with TPU having a direct correlation between microplastics and infill percent while PLA had an inverse relationship between microplastics and infill percent. These findings support a more nuanced view of microplastics that seep into water from 3D printed plastics as the confounding factors can have different results based on the type of polymer printed.

The gap in research regarding microplastics in water resulting from 3D printed plastics has been thoroughly investigated through the lens of plastic polymer, infill percent, and temperature among other factors leading to a potential trend that could be developed in further testing: an inverse correlation between particulate matter emissions while printing and microplastics observed in water post-print. This comes as a result of certain polymers emitting greater particulate emissions but still having limited microplastics in the sampled waters.

All in all, this study has found that 3D printing food safe containers is not as simple as choosing the right filament, but instead a fusion of many assorted factors that decide whether a container will be considered food safe from the standpoint of microplastics. Furthermore, it is unlikely that 3D printing will establish itself as a suitable manufacturing method for food containers because of the large numbers of observed microplastics in the water samples.

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